

PRACTICAL COURSE WORKBOOK

Applied Prompt Engineering

Build a responsible AI workflow for a real work task

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a documented prompt, verification and revision workflow
SKILLS	Prompting, Chain-of-thought, RAG, Critical evaluation, Documentation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use Prompting appropriately in a realistic, bounded task.
- Use Chain-of-thought appropriately in a realistic, bounded task.
- Use RAG appropriately in a realistic, bounded task.

WHO THIS IS FOR

People who want dependable, responsible AI-assisted workflows.

BEFORE YOU BEGIN

- Basic computer and web skills

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Applied Prompt Engineering is a structured, practical course covering Prompting, Chain-of-thought, RAG. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Prompting: Foundations	1. Capabilities with Prompting 2. Limitations with Chain-of-thought 3. Responsible use with RAG
02 Chain-of-thought: Working methods	1. Clear instructions with Prompting 2. Context design with Chain-of-thought 3. Verification with RAG
03 RAG: Applied workflows	1. Research with Prompting 2. Creation with Chain-of-thought 3. Automation with RAG
04 Prompting: Quality and review	1. Evaluation with Prompting 2. Privacy with Chain-of-thought 3. Repeatable practice with RAG

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

Prompting: Foundations

Apply Prompting through capabilities, limitations, responsible use.

LESSON 01 / 18 MIN

Capabilities with Prompting

Learning objectives

- Explain capabilities in the context of Applied Prompt Engineering.
- Apply Prompting to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

capabilities, Prompting, ai-tools

Retrieval prompt

In Applied Prompt Engineering, which evidence best supports a capabilities result produced with Prompting?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 02 / 22 MIN

Limitations with Chain-of-thought

Learning objectives

- Explain limitations in the context of Applied Prompt Engineering.
- Apply Chain-of-thought to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

limitations, Chain-of-thought, ai-tools

Retrieval prompt

In Applied Prompt Engineering, which evidence best supports a limitations result produced with Chain-of-thought?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 03 / 26 MIN

Responsible use with RAG

Learning objectives

- Explain responsible use in the context of Applied Prompt Engineering.
- Apply RAG to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

responsible use, RAG, ai-tools

Retrieval prompt

In Applied Prompt Engineering, which evidence best supports a responsible use result produced with RAG?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

Chain-of-thought: Working methods

Apply Chain-of-thought through clear instructions, context design, verification.

LESSON 04 / 30 MIN

Clear instructions with Prompting

Learning objectives

- Explain clear instructions in the context of Applied Prompt Engineering.
- Apply Prompting to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

clear instructions, Prompting, ai-tools

Retrieval prompt

In Applied Prompt Engineering, which evidence best supports a clear instructions result produced with Prompting?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 05 / 18 MIN

Context design with Chain-of-thought

Learning objectives

- Explain context design in the context of Applied Prompt Engineering.
- Apply Chain-of-thought to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

context design, Chain-of-thought, ai-tools

Retrieval prompt

In Applied Prompt Engineering, which evidence best supports a context design result produced with Chain-of-thought?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 06 / 22 MIN

Verification with RAG

Learning objectives

- Explain verification in the context of Applied Prompt Engineering.
- Apply RAG to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

verification, RAG, ai-tools

Retrieval prompt

In Applied Prompt Engineering, which evidence best supports a verification result produced with RAG?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

RAG: Applied workflows

Apply RAG through research, creation, automation.

LESSON 07 / 26 MIN

Research with Prompting

Learning objectives

- Explain research in the context of Applied Prompt Engineering.
- Apply Prompting to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

research, Prompting, ai-tools

Retrieval prompt

In Applied Prompt Engineering, which evidence best supports a research result produced with Prompting?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 08 / 30 MIN

Creation with Chain-of-thought

Learning objectives

- Explain creation in the context of Applied Prompt Engineering.
- Apply Chain-of-thought to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

creation, Chain-of-thought, ai-tools

Retrieval prompt

In Applied Prompt Engineering, which evidence best supports a creation result produced with Chain-of-thought?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 09 / 18 MIN

Automation with RAG

Learning objectives

- Explain automation in the context of Applied Prompt Engineering.
- Apply RAG to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

automation, RAG, ai-tools

Retrieval prompt

In Applied Prompt Engineering, which evidence best supports a automation result produced with RAG?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

Prompting: Quality and review

Apply Prompting through evaluation, privacy, repeatable practice.

LESSON 10 / 22 MIN

Evaluation with Prompting

Learning objectives

- Explain evaluation in the context of Applied Prompt Engineering.
- Apply Prompting to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

evaluation, Prompting, ai-tools

Retrieval prompt

In Applied Prompt Engineering, which evidence best supports a evaluation result produced with Prompting?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 11 / 26 MIN

Privacy with Chain-of-thought

Learning objectives

- Explain privacy in the context of Applied Prompt Engineering.
- Apply Chain-of-thought to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

privacy, Chain-of-thought, ai-tools

Retrieval prompt

In Applied Prompt Engineering, which evidence best supports a privacy result produced with Chain-of-thought?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 12 / 30 MIN

Repeatable practice with RAG

Learning objectives

- Explain repeatable practice in the context of Applied Prompt Engineering.
- Apply RAG to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

repeatable practice, RAG, ai-tools

Retrieval prompt

In Applied Prompt Engineering, which evidence best supports a repeatable practice result produced with RAG?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable Applied Prompt Engineering project using Prompting, Chain-of-thought, RAG.

DELIVERABLE	A working Applied Prompt Engineering artefact plus an evidence-based self-review.
TOOLS	A suitable Prompting environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using Prompting.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER

State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS

List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

AI Risk Management Framework 1.0

NIST - reviewed 2026-08-21

<https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-ai-rmf-10>

AI Principles

OECD - reviewed 2026-08-21

<https://www.oecd.org/en/topics/ai-principles.html>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Prompting scope.